

Stability of Floquet sidebands and quantum coherence in one-dimensional strongly interacting spinless fermions

Karun Gadge^{*} and Salvatore R. Manmana[†]

Institute for Theoretical Physics, Georg-August-University Göttingen, Friedrich-Hund-Platz 1, D-37077 Göttingen, Germany



(Received 1 April 2025; revised 5 August 2025; accepted 7 August 2025; published 25 August 2025)

For strongly correlated quantum systems, fundamental questions about the formation and stability of Floquet-Bloch sidebands (FBs) upon periodic driving remain unresolved. Here, we investigate the impact of electron-electron interactions and perturbations in the coherence of the driving on the lifetime of FBs by directly computing time-dependent single-particle spectral functions using exact diagonalization (ED) and matrix product states (MPS). We study interacting metallic and correlated insulating phases in a chain of correlated spinless fermions. At high-frequency driving we obtain clearly separated, long-lived FBs of the full many-body excitation continuum. However, if there is significant overlap of the features, which is more probable in the low-frequency regime, the interactions lead to strong heating, which results in a significant loss of quantum coherence and of the FBs. Similar suppression of FBs is obtained in the presence of noise. The emerging picture is further elucidated by the behavior of real-space single-particle propagators, of the energy gain, and of the momentum distribution function, which is related to a quantum Fisher information that is directly accessible by spectroscopic measurements.

DOI: [10.1103/PhysRevB.112.085144](https://doi.org/10.1103/PhysRevB.112.085144)

I. INTRODUCTION

A promising direction of research for realizing interesting quantum states of matter is Floquet engineering, where the interaction of the light field with electrons in materials [1–7] is used to tailor the band structure. For example, such periodically driven systems are predicted to offer a tunable platform to realize Fractional Chern insulators (FCIs) [8,9], engineered topological states [10–15], and coherent excitations in experiments with ultracold gases [9,16–20] (for a review see Ref. [21]). A hallmark of such periodically driven crystalline lattice structures is the emergence of Floquet-Bloch sidebands (FBs). These FBs are replica of the original band structure and appear at energies which are integer multiples of the driving frequency. They can directly be detected in pump-probe experiments by using time- and angle-resolved photoemission spectroscopy (trARPES) [22–30], which gives insight into the time-dependent single-particle spectral function. The direct observation of FBs has been reported, e.g., for the topological insulator Bi_2Se_3 [23,24], for the semiconductor material WSe_2 [26], and recently for monolayer graphene [27,28] and for the topological antiferromagnet MnBi_2Te_4 [29]. These experimental findings are for weakly interacting systems. Model calculations and experimental work shows that it is nontrivial to understand the Floquet

physics even for weak interactions [26,31,32]. Theoretical studies of interacting systems have been performed using cluster perturbation theory, or in infinite spatial dimensions using Floquet-DMFT and out-of-equilibrium DMFT [33–36]. However, in 1D and 2D strongly interacting systems the interplay of Floquet driving and many-body effects on the spectral functions and on the stability of FBs is still a topic of ongoing investigations [26].

Floquet systems have also been explored in the context of many-body localization (MBL) and thermalization [39–43]. In particular, by fine-tuning drive parameters or suppressing hopping terms, one can induce localization even in clean systems (dynamical localization [37,44]), revealing rich regimes of Floquet MBL and prethermal dynamics. Other important effects of periodic driving are energy absorption [34,35,45], dressing of electrons [37], light induced gaps [46,47], metal-insulator transitions [34–36], and in strongly correlated systems it can also induce in-gap features in the presence of symmetry broken ground states [38]. A major challenge in interacting Floquet systems is the significant heating [43,48–55], which drives the system toward an infinite-temperature state. As a result, one expects Floquet effects to play a role on some transient time scale, before the heating wins [48].

The scope of this paper is to investigate how for strongly correlated electron systems the interactions, the driving parameters, and noise influence the Floquet physics. We do so by directly studying the time evolution of the retarded time-dependent spectral function $A_k^{\text{ret}}(t, \omega)$, of real space correlations, and of the momentum distribution function $\langle n_k \rangle(t)$, which relates to a Quantum Fisher Information (QFI) [56]. All results are obtained with time-dependent matrix product states (MPS) [57,58] and exact diagonalization (ED) [59–61] techniques. While most of the calculations can be done using ED,

^{*}Contact author: karun.gadge@uni-goettingen.de

[†]Contact author: salvatore.manmana@uni-goettingen.de

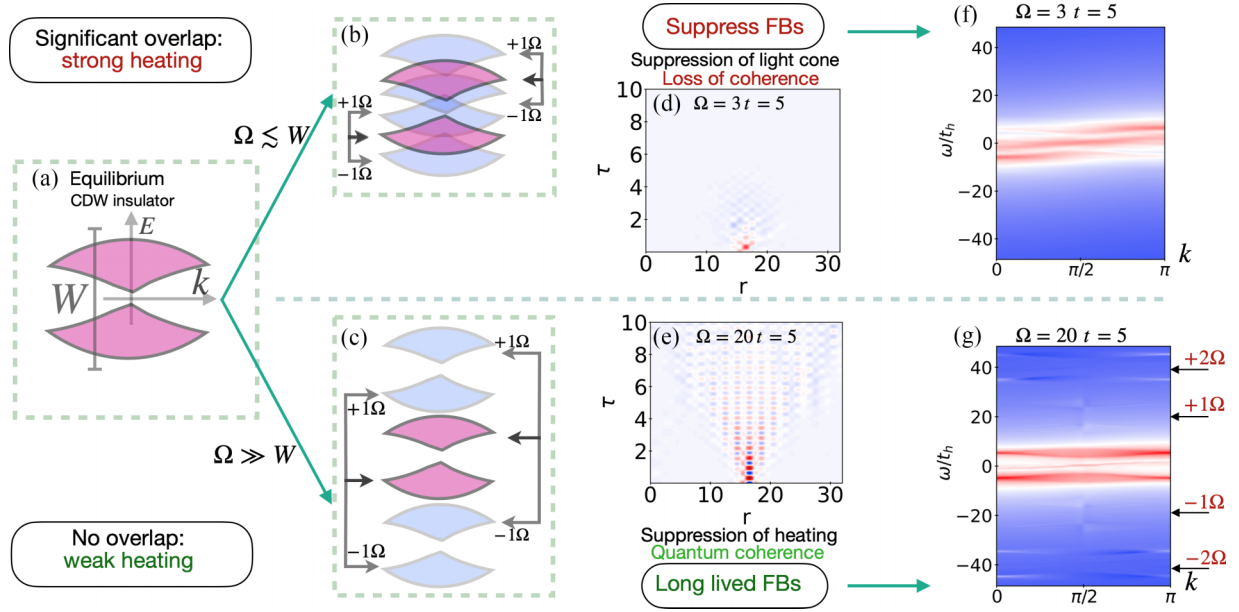


FIG. 1. Summary of the two extreme cases of our findings. (a) Sketch of a typical equilibrium result for the retarded spectral function $A_k^{\text{ret}}(\omega)$ in the charge density wave (CDW) insulating phase at half filling for model (1). We denote by W the *spectral width* (see Sec. III), in which the spectral function has populated continua. (b) and (c) illustrate the general expectation for FBs in such a strongly correlated system: In both sketches, the magenta region illustrates the zeroth Floquet sector, which is located at the position of the original equilibrium spectral function. Note that, due to the driving, in comparison to the equilibrium result the shape can be modified [37], and additional effects (e.g., in-gap bands [38]) can come into appearance. This spectral function in the zeroth Floquet sector then is replicated at energies, which are multiples of the driving frequency $\pm n\Omega$. Here, we only sketch the first replicas at $\pm\Omega$ (light blue). In (b) we sketch the situation, when the driving frequency $\Omega < W$. This leads to substantial overlap of the different Floquet sectors. In contrast to this, in (c) we sketch the situation if $\Omega > W$, so that there is no overlap between the Floquet sectors. (d) and (e) illustrate the effect of the driving on correlations in real space. They display the time evolution of the single particle propagator Eq. (5) in real space in the CDW phase for driving frequencies $\Omega = 3.0$ and $\Omega = 20.0$, respectively, at waiting time $t = 5$, indicating the stability of quantum coherence in both regimes. (f) and (g) show examples for the resulting time dependent spectral function $A_k^{\text{ret}}(t, \omega)$ [Eq. (4)] in the CDW state at low- and high-frequency driving, $\Omega = 3.0$ (no FBs visible) and $\Omega = 20$ (FBs visible at $\pm n\Omega$ indicated by the arrows), respectively, at waiting time $t = 5$. All results are at $V/t_h = 5$.

it is useful to compare the results for the spectral functions with MPS results, since we can treat larger systems and hence reach a better resolution in k -space there. Studying larger systems will also be beneficial when exploring the dynamics of real-space correlations (see Sec. III D). We focus on one-dimensional (1D) systems, as MPS methods are particularly efficient in this setting. Since correlation effects are strongest in 1D, we believe that studying them in this context will provide a generic understanding that is likely to hold in higher dimensions as well. This, in turn, will help us understand how interactions affect the stability of FBs in correlated systems in general. We complement this by studying coherence in real space correlations and in the momentum distribution function, which is a simple QFI, in strongly interacting Floquet-driven systems. This is motivated by recent inelastic neutron scattering studies on quantum magnets, which analyze the time evolution of correlation functions in real space and of the QFI [62–65]. Further work relating the single-particle spectral function [65,66] in- and out-of-equilibrium [67,68] to QFIs is ongoing.

A summary showing the two extreme cases with and without FBs is sketched in Fig. 1. We find in the high-frequency regime that the FBs are stable for long times even in the presence of strong interactions, and are only suppressed when adding noise to the drive. In the low-frequency regime, they

are suppressed if there is a large overlap between the FBs and the original spectral function. In general, our results for clean monochromatic driving indicate that for interacting systems the stability of the FBs depends on this overlap.

This paper is organized as follows. In Sec. II, we introduce the model used in our study. Our main findings are presented in Sec. III, where we discuss the system's behavior under a low-frequency drive (Sec. III A), a high-frequency drive (Sec. III B), the stability of the FBs against noise (Sec. III C), and the evolution of real-space correlations, the momentum distribution function, and heating (Sec. III D). We summarize our conclusions and provide an outlook in Sec. IV. Several technical details and complementary results are provided in the Appendixes. Details on the numerical methods are provided in Appendixes A and B, with Appendix B discussing details of the Fourier transform used, in particular the choice of a window function. Further supporting results for the main text and an error analysis is provided in Appendixes C, D, E, and F.

II. MODEL

We consider a system of periodically driven one-dimensional interacting spinless fermions (tV -chain),

$$H(t) = -t_h \sum_j (e^{iA_V(t)} c_j^\dagger c_{j+1} + \text{H.c.}) + V(n_j n_{j+1}). \quad (1)$$

Here, $c_j^{(\dagger)}$ is the annihilation (creation) operator for a spinless fermion at site j , $n_j = c_j^\dagger c_j$ is the operator of the local density on site j , t_h the hopping strength between nearest neighboring lattice sites, and V the strength of the interaction between particles sitting on neighboring sites. The periodic driving is realized as Peierls' substitution [69] via a time-dependent vector potential $A_V(t) = A_0 \sin(\Omega t)$, which is activated at time $t = 0$. Unless stated otherwise, we use open boundary conditions (OBC) and the hopping parameter is set to $t_h \equiv 1$ throughout, as well as $\hbar \equiv 1$. Without the driving (i.e., for $A_0 = 0$), the model Eq. (1) at half filling exhibits a Berezinskii-Kosterlitz-Thouless-like (BKT) phase transition from a Luttinger liquid (LL) [70] to a strongly correlated charge density wave (CDW) insulator at $V/t_h = 2$, as obtained from Bethe ansatz after mapping to an XXZ spin-chain via Jorgan-Wigner transform [71]. In order to suppress artifacts due to the OBC, we apply a pinning-field $H_{\text{pin}} = \mu n_j$ at one of the edge sites, which enforces the numerics to converge to one of the two possible CDW ground states [38,72]. Throughout the paper, the initial state is prepared as a ground state of the system at temperature $T = 0$. We compute the nonequilibrium generalization of the spectral function by performing a Fourier transform (FT) of the single-particle Green's function,

$$G_{\alpha\beta}^{\text{ret}}(t, t') = -i\theta(t - t')\{c_\alpha(t), c_\beta^\dagger(t')\}. \quad (2)$$

The FT to ω -space is not unique [73]. Typical choices are Wigner coordinates, or relative-time coordinates. The qualitative behavior typically does not depend on the particular choice of these coordinates [73–75]. For our numerical approach, relative-time coordinates $\tau = t' - t$ for the FT are more convenient, since in this way one can reduce the number of MPS states that need to be stored in the course of the calculation. The protocol is

$$\begin{aligned} |\psi\rangle &\xrightarrow[t=0: \text{turn on driving}]{\text{evolve in time variable } t} |\psi(t)\rangle \\ |\psi(t)\rangle &\xrightarrow[\text{compute Green's fct.}]{\text{evolve in } \tau} G_{\alpha,\beta}(t, t + \tau) \\ G_{\alpha,\beta}(t, t + \tau) &\xrightarrow[\tau \rightarrow \omega]{\text{FT}} A_k(t, \omega), \end{aligned} \quad (3)$$

where we interpret the evolution time t as a waiting time, since we let evolve the state for time t under the influence of the periodically driven time-dependent Hamiltonian before we compute the Green's function [38,72]. The evolution in the time variable τ is also done with the time-dependent Hamiltonian, and then the retarded single-particle spectral function is obtained via a FT over τ ,

$$A_k^{\text{ret}}(t, \omega) = -\text{Im} \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} d\tau e^{i\omega\tau} W(\tau) G_{kk}^{\text{ret}}(t, t + \tau). \quad (4)$$

The function $W(\tau)$ is a suitable window function (see Appendix B). Note that even at waiting time $t = 0$ the results differ from the ground state behavior, since the Fourier-transform to ω -space is done on results, which were obtained with the time-dependent Hamiltonian (1). The retarded spectral function Eq. (4) can then be interpreted as the spectral function measured at waiting time t , with the driving being turned on at $t = 0$. In contrast to the equilibrium case,

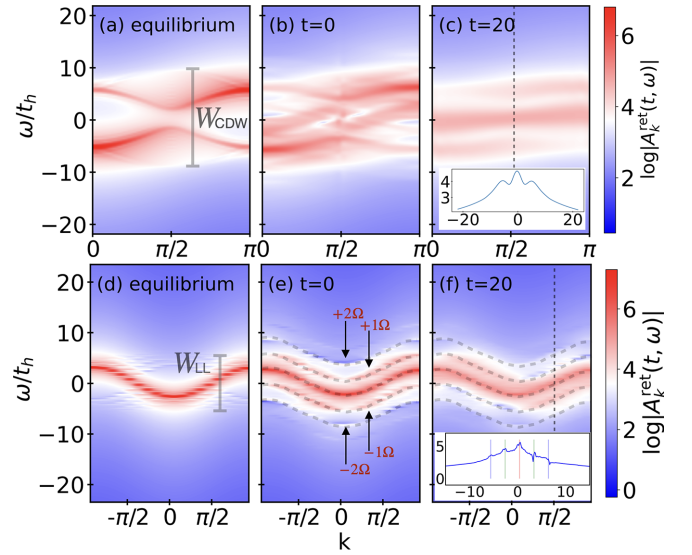


FIG. 2. Retarded single-particle spectral function $A_k^{\text{ret}}(t, \omega)$ [Eq. (4)] for the tV -chain (1). (a)–(c): Results in the CDW phase for $V/t_h = 5$ for $L = 32$ and OBC at half filling obtained with MPS. (a) Results at equilibrium. $W_{\text{CDW}} \approx 18$ estimates the spectral width, in which $A_k^{\text{ret}}(\omega) > \epsilon$, where ϵ is 2% of the peak value of $A_k^{\text{ret}}(t, \omega)$ (see main text). (b) Driven case with $\Omega = 3$ and $A_0 = 1.0$ at waiting time $t = 0$. (c) The same driven case at waiting time $t = 20$. The inset in (c) shows the result at fixed $k = \pi/2$ for $t = 20$. (d)–(f): Results in the LL phase $V/t_h = 1.5$ for $L = 18$ and PBC at half filling obtained with ED. (d) Results at equilibrium, $W_{\text{LL}} \approx 10$ denotes the spectral width, as before. (e) and (f) nonequilibrium results for the same driving parameters as in (b) and (c). In (e) the first two side bands are indicated with arrows. The inset in (f) shows the results at fixed $k = \pi/2$ for $t = 20$, vertical lines indicate the position of the side bands. Dashed lines in (e) and (f) are guides to the eye for locating the side bands.

$A_k^{\text{ret}}(t, \omega)$ can have negative values, see, e.g., Refs. [76,77], which, however, in some strongly interacting systems have been found to be absent at later times [38,77]. Here, in the context of the existence of FBs, we focus our discussion on the absolute value $|A_k^{\text{ret}}(t, \omega)|$. We use OBC for our MPS results, since MPS are more efficient there [57]. For the results at long times, we need to use Lanczos time evolution [78], for which we use PBC and smaller system sizes. The results with OBC and PBC show the same behavior, so that in the main part of the paper we discuss them side by side, where needed. For details of the MPS calculations and an error estimate, see Appendixes A, D, and E.

III. RESULTS

We present in Figs. 2 and 3 our main results. In Fig. 2, we investigate the dynamics at low-frequency driving for two distinct phases of the system: the CDW insulator (top row) and the LL phase (bottom row). For the high-frequency drive, we focus on the CDW insulator, since here interaction effects would be strongest, see Fig. 3 for our results.

In order to better distinguish between the low- and high-frequency regime, we introduce the spectral width W in the ground-state retarded spectral function at equilibrium,

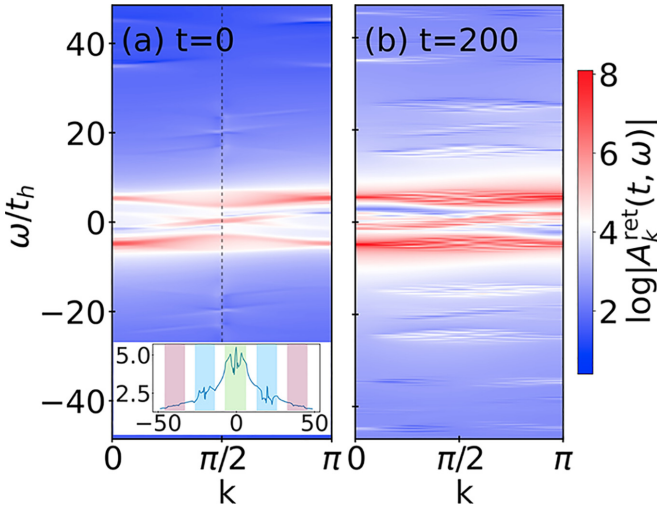


FIG. 3. The same as in Fig. 2 for $V/t_h = 5$, but at high-frequency driving $\Omega = 20$ and driving amplitude $A_0 = 1.6$. (a) shows MPS results for $L = 32$ and OBC, while the long-time results in (b) are obtained using Lanczos time evolution with $L = 18$ sites and PBC. The inset in (a) shows the results at fixed $k = \pi/2$; the green highlighted region indicates the spectral function in the zeroth Floquet sector, the blue shaded regions indicate the first FBs, and the magenta shaded regions indicate the second FBs.

$A_k^{\text{ret,eq}}(\omega)$. In the context of our discussion, we are most interested in the influence of correlation effects on the properties of the FBs. On general grounds, for noninteracting systems we expect well-defined band structures and no continua in $A_k^{\text{ret,eq}}(\omega)$. Hence, the presence of continua in $A_k^{\text{ret,eq}}(\omega)$ is indicative for correlation effects. This is further confirmed by mean-field calculations (Hartree-Fock), which cannot reproduce the excitation continua discussed here in the tV-chain, see Ref. [38]. In our estimate of W , we therefore focus on the extend in ω -direction covered by such continua. In Fig. 2(a) we denote by W_{CDW} the spectral width in the CDW phase; since for these parameters the most prominent aspects of the spectral function are continua, W_{CDW} essentially covers the entire region in which $|A_k^{\text{ret,eq}}(\omega)| > \epsilon$, with ϵ a suitable threshold value. Note that the support of $A_k^{\text{ret,eq}}(\omega)$ can extend over all ω -values [79]. In general, the continua may decay as a power law and hence have finite weight over a wide range of ω . However, for the following discussion it is only important to distinguish highly populated regions from weakly populated regions, so that choosing ϵ small enough suffices. Here, we choose ϵ such that $|A_k^{\text{ret,eq}}(\omega)|$ is larger than 2% of the peak value, leading to a value $W_{\text{CDW}} \approx 18$. In the LL case shown in Fig. 2(d), the situation is different. Here, $A_k^{\text{ret,eq}}(\omega)$ is dominated by a cos-like feature, plus additional weak excitation continua [79]. Using again a threshold value of 2% of the peak value, we obtain a spectral width $W_{\text{LL}} \approx 10$. Based on these considerations, we identify $\Omega/t_h < W$ as a low-frequency drive and $\Omega/t_h > W$ as a high-frequency drive.

In more detail: in Fig. 2(a) we present MPS results for $A_k^{\text{ret,eq}}(\omega)$ deep in the CDW-regime at $V/t_h = 5$ for $L = 32$ at half filling. Two continua of holon excitations, separated by a gap, are observed within the energy range $\omega \in [-10, 10]$, consistent with previous studies [38]. The gap size Δ can be

calculated using the Bethe ansatz, which for $V/t_h = 5$ obtains $\Delta/t_h \approx 1.576$ [71]. This is in agreement with the gap size seen in Fig. 2(a), which is affected by finite-size effects and the limited spectral resolution of our methods. Figure 2(d) shows ED results for $A_k^{\text{ret,eq}}(\omega)$ for $L = 18$ at half filling for $V/t_h = 1.5$, i.e., in the LL phase, which shows the aforementioned cosine-like dispersion with a buildup of a weakly populated excitation continuum without gap. This weakly populated region is observed within the energy range $\omega \in [-5, 5]$, leading to our estimated value of W_{LL} .

The equilibrium results shown in Figs. 2(a) and 2(d) serve as the starting point for our investigation. A key aspect to consider is the potential overlap of the Floquet replicas with the original GS spectral function: in case of a substantial overlap, interference or scattering events seem likely, which should lead to heating. This is expected for $\Omega \lesssim W$.

In driven systems, heating can be suppressed by various mechanisms. For example, in disordered systems, Floquet many body localization (MBL) has been discussed as a possibility to avoid heating up of the system to infinite temperatures, at least on transient time scales [39,41,80–82]. Without disorder, it has been reported in the high frequency case that heating is suppressed due to an emergent effective static Hamiltonian. A transient prethermal state forms, which can be long lived, before the system finally heats up to infinite temperatures [50,52,54,55,83]. Indeed, our findings for high frequency driving discussed in Sec. III B indicate this behavior. In such systems, the driving frequency is typically compared to the intrinsic energy scales of the system [52], and in some cases, to the noninteracting bandwidth [50]. At low frequencies, the picture of an effective static Hamiltonian, which is obtained by perturbative ansatzes, may lose its validity. This makes the question of heating in a generic many-body system a more difficult one due to nonuniversal heating rates in low-to-intermediate driving frequencies [49] and may also depend on the presence of high-energy continuum states [52].

Our findings indicate that the suppression of heating is related to the overlap of the Floquet replica, and also to the actual weight of the spectral function and its replica in the overlapping regions. We use the spectral width W as defined above in order to estimate the role of correlation effects. This allows us to determine the relevant energy scale against which the driving frequency is compared, as illustrated in Fig. 1(a). However, it is not clear *a priori* whether in the case of overlapping Floquet replicas a transient time scale remains, on which FBs can form. This can be contrasted to noninteracting systems, where we do not expect heating due to the absence of scattering terms, even in the presence of a complicated band structure (e.g., multiple bands).

A. Low-frequency drive

Here, we set the driving amplitude $A_0 = 1.0$ and driving frequency $\Omega/t_h = 3.0$, which is smaller than $W_{\text{CDW}} \approx 18$ and smaller than $W_{\text{LL}} \approx 10$. The results in the CDW phase with $V/t_h = 5.0$ are shown in Figs. 2(a)–2(c). One would expect replicas of the entire equilibrium spectral function, which, due to the small value of Ω/t_h , would significantly overlap with the original signal and with each other. In Fig. 2(b) we observe a significant spectral weight within the gap

region, but no signature of FBs. The in-gap feature at $t = 0$ exhibits a rich structure, which is reminiscent to the avoided-crossing scenario expected in noninteracting two-band models [15,31,34,35,84]. However, as shown in Fig. 2(c) for $t = 20.0$, these features vanish at later times, and a broad continuum is obtained, with a stronger weight inside the original gap region. This is further illustrated in the inset of the Fig. 2(c) for momenta $k = \pi/2$ (dashed line). Further results at different momenta k show similar behavior and are shown for $t = 0$ and $t = 20$ in Appendix F. This melting of the gap features hints towards strong heating effects due to the chosen drive parameters, see further below. This indicates that once the replicas overlap in this strongly interacting system, heating prevails and the spectral function does not show FBs.

In comparison to the gapped CDW phase, in the gapless LL phase also at $\Omega/t_h = 3$ we obtain FBs, which persist also at intermediate times, see Figs. 2(d)–2(f) (results are obtained with Lanczos time evolution, as the low-frequency case exhibits higher entanglement buildup in MPS, leading to lower accuracy. See Appendixes E and D for further discussion): Figure 2(d) shows the spectral function in equilibrium, which follows a cosine-like dispersion with a buildup of a weakly populated excitation continuum without gap. In Fig. 2(e), we observe an overlap of this weakly populated excitation continuum and the FBs at $t = 0$. The FBs are clearly visible, even at $t = 20$ in Fig. 2(f). This is further illustrated in the inset of Fig. 2(f) for $k = \pi/2$. However, the overlap of the FBs and the original spectral function now leads to a weaker signal of the FBs compared to the $t = 0$ case. In addition, the intensity of the FBs is time dependent, and at longer waiting times $t \gtrsim 50/t_h$ they will be suppressed (see Appendix C). Hence, our results indicate that in both, the interacting LL and in the CDW phases, overlapping bands are the main cause for the suppression of FBs due to the enhancement of scattering and the resulting heating. However, if the overlap is weak as in our results for the LL case, the FBs persist on a transient time scale, which for our parameters is $t_{\text{transient}} \sim 50/t_h$ (see Appendix C for additional intermediate and longer waiting time instances).

B. High-frequency drive

In Fig. 3 we present results for the same setup, but with $A_0 = 1.6$, $\Omega/t_h = 20 > W_{\text{CDW}}$, and $V/t_h = 5.0$. At waiting time $t = 0.0$ in Fig. 3(a), we observe clear FBs at integer multiples of $\pm\Omega$. These FBs exhibit the full excitation continuum with renormalized bandwidth (since $t_h^{\text{eff}} = J_0(A_0)t_h$ [37,45,48]) and the emergence of an in-gap band, see Ref. [38]. This shows that also in the strongly interacting case the FBs are replicas of the full spectral function of the zeroth Floquet sector, including all features. This is further illustrated in the inset of Fig. 3(a) for $k = \pi/2$ (dashed line); see Appendix F for further cross sections at other values of k . The question arises, if these FBs are stable in time, or if scattering between the particles eventually will destroy coherence, leading to a suppression of the FBs in $A_k^{\text{ret}}(t, \omega)$ at later times. A simple estimate from perturbation theory suggests scattering becomes visible on time scales $\sim 1/V^2$. Hence, a suppression of the FBs at waiting times $t \lesssim 1$ would follow. However, in Fig. 3(b), we reach longer waiting times using

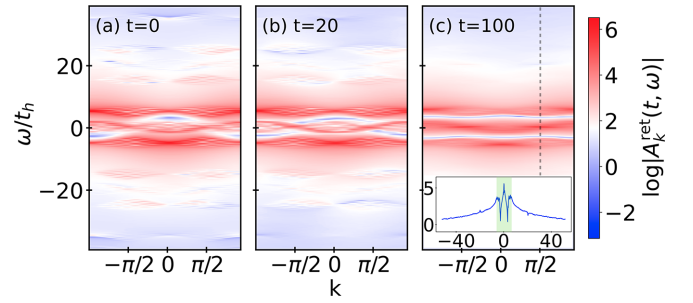


FIG. 4. The same as in Fig. 3, but in the presence of time-dependent noise in the driving, where $\Omega = 20 \pm \Omega_{\text{noise}}$ and $\Omega_{\text{noise}} \in [0, 0.1]$. The results are obtained using Lanczos time evolution with $L = 18$ and PBC. Panels (a)–(c) show $A_k^{\text{ret}}(t, \omega)$ at waiting times $t = 0$, $t = 20$, and $t = 100$, respectively. The inset in (c) shows results at fixed $k = \pi/2$ at $t = 100$. All results show an average over eight realizations of the random noise.

the Lanczos time evolution, we have to restrict to system sizes of $L = 18$ (PBC). Our results indicate that even at these substantially longer times of $t = 200$, the FBs remain stable. In contrast to low frequency drive where heating suppresses FBs already at short times $t \sim 5$ (see above). This demonstrates that high-frequency driving can yield FBs even in strongly interacting quantum many-body systems, exhibiting stability over the extended timescales we examined.

C. Stability of the FBs against noise

We now test these findings by explicitly breaking the coherence when adding incoherent noise to the drive and ask for the stability of the FBs as a function of time. In order to reach longer times, we again apply the Lanczos time evolution method for $L = 18$ (PBC) at half filling, $V/t_h = 5.0$, $A_0 = 1.6$, and a frequency $\Omega = 20 \pm \Omega_{\text{noise}}$, where $\Omega_{\text{noise}} \in [0, 0.1]$ represents the random time-dependent noise in the driving frequency, such that $A_V(t) = A_0 \sin((\Omega \pm \Omega_{\text{noise}})t)$. We discretize the time variable such that $t = ndt$, with $n \in \mathbb{N}_0$, and then compute $A_k^{\text{ret}}(t, \omega)$ when applying $A_V(t) = A_0 \sin((\Omega \pm \Omega_{\text{noise}})ndt)$. Note that the results depend on the value of dt , as discussed in Appendix E. Here, we present results for $dt = 0.005$ in Fig. 4. Figures 4(a) and 4(b) demonstrate that, at small to intermediate waiting times, despite the introduction of noise, FBs are stable. However, at $t = 20$ in Fig. 4(b), we observe a noticeable suppression of the second-order FBs. In Fig. 4(c), for waiting time $t = 100$, this suppression becomes even more pronounced, with the cross section at $k = \pi/2$ shown in the inset of Fig. 4(c) clearly showing the suppression of all FBs. Additionally, we notice a higher intensity inside the gap region, similar to what we found for low frequency driving.

D. Real space correlations, momentum distribution function, and heating

We complement our results for $A_k^{\text{ret}}(t, \omega)$ by studying the time evolution of correlation functions in real space and of the momentum distribution function $\langle n_k \rangle(t)$, accompanied by the behavior of the energy as a function of time. This helps us to monitor the heating, which is intrinsic to driven

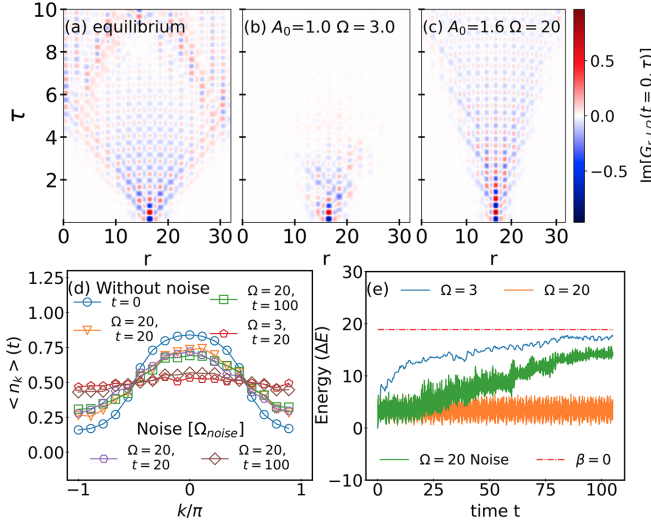


FIG. 5. Results at $V/t_h = 5$: (a)–(c) show the imaginary part of $G_{r,L/2}(t, \tau)$ [Eq. (5)] (MPS, $L = 32$, OBC) for the equilibrium case, and for the driving parameters of Figs. 2 and 3 at waiting time $t = 0$. (d) shows $\langle n_k \rangle(t)$ at $\Omega = 3$ and at $\Omega = 20$, with and without noise for $L = 18$ and PBC. (e) shows typical time evolution of the energy gain compared to the initial state (ground state energy E_{GS}), $\Delta E(t) = \langle H(t) \rangle - E_{GS}$ for different driving frequencies for $L = 14$ and PBC.

interacting systems [49,50,52–55]. In the following we focus on the case $V/t_h = 5.0$. We apply our low and high frequency driving protocols in Figs. 5(a)–5(c) to the imaginary part of the real-space single-particle propagator

$$G_{r,L/2}(t, \tau) = \langle \psi(t) | [c_r^\dagger(\tau), c_{L/2}] | \psi(t) \rangle. \quad (5)$$

Figure 5(a) shows the typical light-cone-like behavior expected for gapped ground states [85–92]. Additionally, one observes a ‘checkerboard-like’ pattern, which is similar to the one observed for spinons in spin correlation function in neutron scattering experiments [62]. Here, we expect this should translate to the presence of holon excitations. In stark contrast to the equilibrium results, we see in Fig. 5(b) that for low frequency driving already at $t = 0$ the correlation function is quickly suppressed as a function of time. This demonstrates that in this case coherence is quickly lost, which suggests the absence of FBs in $A_k^{\text{ret}}(t, \omega)$. However, for the high-frequency driving case shown in Fig. 5(c), again a light-cone is visible, which is narrow due to the reduced hopping $t_h^{\text{eff}} = J_0(A_0)t_h$, which leads to a lower propagation velocity of the perturbations in the system. In the equilibrium case one observes ‘wakes’ [62], which are created inside the light cone. In the driven case, they are more pronounced at later times, and every second one seems to be suppressed. Also, the checkerboard pattern is very similar to the one at equilibrium, but with slightly slower oscillation frequency in time. This is also obtained at later waiting times (see Appendix D). These findings illustrate that coherence is maintained, suggesting that here long-lived FBs are possible, as indeed found in Fig. 3.

Coherence effects can also be studied in the momentum distribution function $\langle n_k \rangle(t) = \langle c_k^\dagger c_k \rangle(t)$, with $c_k^{(\dagger)} = 1/\sqrt{L} \sum_{r=1}^L e^{(-)ik \cdot r} c_r^{(\dagger)}$, which is accessible via time-of-flight

experiments in cold-gases setups [16]. In Fig. 5(d) we show several snapshots of $\langle n_k \rangle(t)$ for $\Omega = 3.0$ and $\Omega = 20.0$ with and without noise (ED results for $L = 18$ and $V/t_h = 5$). At equilibrium, we obtain the typical result for a strongly correlated insulator. This behavior persists in the high frequency regime up to the longest times studied by us, here $t = 100$. However, when turning on the noise, or in the low frequency regime, all features are lost and $\langle n_k \rangle(t) \approx \text{const}$ for all values of k . In the low frequency case, this happens already at short times $t \sim 5$, while in the case with noise it happens only at later times $t \gtrsim 30$. This featureless result for $\langle n_k \rangle$ is expected at very high or infinite temperatures, further showing that in these cases coherence gets lost.

The findings for $G_{r,L/2}(t, \tau)$ and for $\langle n_k \rangle(t)$ can be related to heating effects by comparing with the behavior of the energy gain as a function of time $\Delta E(t) = \langle H(t) \rangle - E_{GS}$ (with E_{GS} the ground state energy), see Fig. 5(e). This has been discussed extensively in the literature [50,52–55]. Further results for $G_{r,L/2}(t, \tau)$, $\langle n_k \rangle(t)$, and $E(t)$ for longer times and at later waiting times, as well as a discussion of the numerical errors, can be found in Appendixes D and E. The results for coherent driving with $\Omega = 3$ and $\Omega = 20$ differ clearly at long times. While for $\Omega = 20$ the energy oscillates around a value, which is clearly not at infinite T , for $\Omega = 3$ it approaches this limit quickly. This seems in agreement with the expected suppression of heating at high driving frequencies [52]. When adding noise, we find intermediate behavior: while at times $t \lesssim 25$ the results for the coherent driving are reproduced, at later times the result at infinite temperatures is approached. This is in agreement with the findings of Ref. [93]. It is interesting that in the presence of noise a clean transient time window exists, in which the behavior is essentially the one of the coherent driving, as can also be seen in Figs. 4(a) and 4(b).

IV. CONCLUSIONS AND OUTLOOK

We directly studied the formation of FBs in the presence of strong correlations in the time-dependent spectral functions for different interaction strengths and driving parameters using MPS and ED methods. Since in the 1D systems studied by us correlation effects are strongest, our results indicate that, in the absence of noise, Floquet engineering should be possible also in strongly correlated systems and for suitable parameters also for low-frequency driving on a transient time scale. This should also hold for strongly correlated materials at low enough temperatures, e.g., cuprates [94]: As long as the coherent driving frequency is larger than the typical spectral width W and noise is suppressed, if the overlap of the FBs and the original spectral function is weak, the FBs persist on a transient time scale, which depends on details of the system. In contrast, for large overlap, we find fast loss of coherence and very strong heating, which leads to an immediate suppression of the FBs. For high frequency driving, (i.e., $\Omega \gg W$), no such overlap is possible, and FBs are long-lived even in the presence of strong interactions. However, incoherent noise in the driving also here leads to a transient timescale after which they are suppressed. This is further corroborated by the behavior of the single-particle propagator $G_{r,L/2}(t, \tau)$ and the momentum distribution function $\langle n_k \rangle(t)$, which is related to a QFI $F_Q(k, t) = \int d\omega A_k^{\text{ret}}(t, \omega) = \langle n_k \rangle(t)$ when

expressing the QFI using the simple entanglement witness operator $\hat{O} = \sum_i (a_i c_i^\dagger + \text{H.c.})$ [56]. This deserves further investigation in order to explore the possibility to relate time-dependent single-particle spectral functions with other QFIs. [66]. Future investigations of FBs in strongly correlated systems should address a more realistic modeling of such materials and include the effect of finite temperatures and of phonons on the stability of the FBs and hence the possibility to realize Floquet engineering in these materials.

ACKNOWLEDGMENTS

We acknowledge interesting discussions with Theo Costi, Alan Tennant, Manuel Buriks, Fabian Heidrich-Meisner, Anatoli Polkovnikov, Michael Schüler, Maksymilian Środa, and the Journal Club of the Stefan Mathias group (in particular Marco Merboldt, Marcel Reutzel, and Stefan Mathias). The calculations were performed on the GOEGrid cluster in Göttingen. We acknowledge financial support by Deutsche Forschungsgemeinschaft (DFG, German Research Foundation) Grants No. 436382789, and No. 493420525, via large equipment grants (GOEGrid), and by the Deutsche Forschungsgemeinschaft (DFG, German Research Foundation) - 217133147/SFB 1073, project B03.

DATA AVAILABILITY

The data that support the findings of this article are openly available [95].

APPENDIX A: DETAILS OF THE CALCULATIONS

We compute

$$G_{kk}^{\text{ret}}(t, \tau) := -i\theta(\tau)\langle\{c_k(t + \tau), c_k^\dagger(t)\}\rangle, \quad (\text{A1})$$

which is the retarded Green's function. For the real-to- k -space transform, in the case of PBC we apply a standard Fourier transform, $c_k^{(\dagger)} = 1/\sqrt{L} \sum_j e^{(-)ijk} c_j^{(\dagger)}$ with momenta $k = 2\pi/L \cdot \{-L/2, \dots, L/2 - 1\}$. For OBC, we apply the sine transform with quasimomenta $k \in \pi/L + 1 \cdot \{1, \dots, L\}$ [38,96–98],

$$c_k = \sqrt{\frac{2}{L+1}} \sum_i \sin(k \cdot i) c_i. \quad (\text{A2})$$

This leads for OBC to a nontrivial time dependence of $\langle n_k \rangle(t)$ even for noninteracting systems, as can be seen as follows. Consider the noninteracting Hamiltonian with Peierls' substitution,

$$H(t) = -t_h \sum_{j=1}^{N-1} (e^{iA_V(t)} c_j^\dagger c_{j+1} + \text{H.c.}). \quad (\text{A3})$$

Using the sine transform, the Hamiltonian is diagonalized, but has time-dependent eigenvalues,

$$H(t) = -t_h \sum_k [\cos(k + A_V(t)) + \cos(k - A_V(t))] c_k^\dagger c_k, \quad (\text{A4})$$

which implies that $\langle n_k \rangle(t)$ is time dependent. (At $t = 0$, this simplifies to the usual expression $H(t = 0) = -2t_h \sum_k \cos(k) c_k^\dagger c_k$.)

The MPS calculations have been carried out using the SymMPS toolkit (developed by Sebastian Paeckel and Thomas Köhler), which is freely available at [99]. For comprehensive reviews we refer to the literature [57,58]. We follow the steps:

$$\begin{aligned} |\psi(t)\rangle &= U_{\text{TDVP}}(t, 0) |\psi_0\rangle, \\ |\psi_I(t)\rangle &= c_I |\psi(t)\rangle, \end{aligned} \quad (\text{A5})$$

$$|\psi_I(t + \tau)\rangle = U_{\text{TDVP}}(t + \tau, t) |\psi_I(t)\rangle,$$

with the time-dependent time evolution operator $U_{\text{TDVP}}(t, t')$, which is implemented using a 2-site time-dependent variational principle (TDVP) [58]. For time-dependent Hamiltonians, there are better approximations (e.g., commutator-free expansions, see Ref. [100]); however, if the value of dt is chosen small enough, we obtain a good accuracy also with this simpler approach. We repeat similar calculations for the operator $c_I^\dagger$. Using the states (A5) we calculate the quantities

$$C_{ml}(t, \tau) = \langle \psi(t + \tau) | c_m^\dagger | \psi_I(t + \tau) \rangle. \quad (\text{A6})$$

We work in units of the energy in which the hopping parameter $t_h \equiv 1$, and $\hbar \equiv 1$ throughout the simulations, which leads to discrete time steps dt in these units. Typically, we set $dt = 0.005$, which is chosen to be one order of magnitude smaller than the highest frequency scale $\sim 1/\Omega$ (with $\Omega = 20$ for the high frequency drive). We vary the bond dimension between 400 and 800. For $\Omega = 20$ for $dt = 0.005$ without noise we observe that the typical discarded weight at the end of the time evolutions is $\lesssim 10^{-8}$, which is a rather small value, indicating a high precision of the results. This is further corroborated by comparing the results for the time evolution of the energy with different values of dt as shown in Fig. 14, where excellent agreement is found (see also the discussion further below). In other cases, in particular for $\Omega = 3$, the discarded weight grows much faster, and we reach rather large maximal values

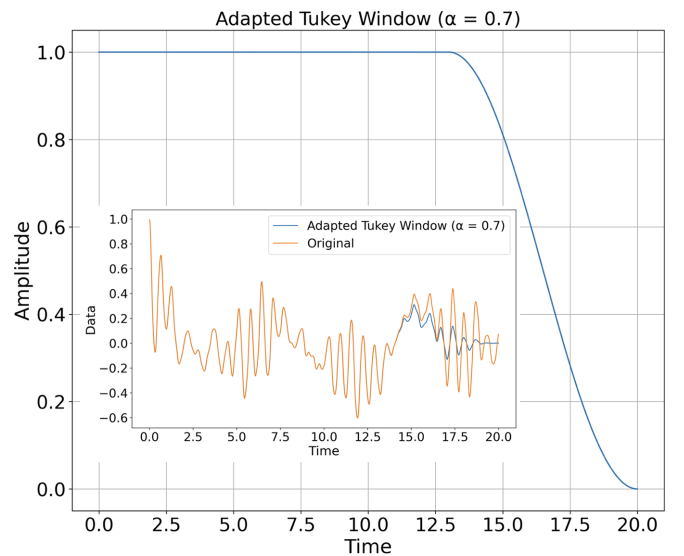


FIG. 6. Damping of the data when applying the adapted Tukey window (B2). The main figure shows the amplitude of the windowing function for $\alpha = 0.7$ and the impact on the raw data is shown in the inset.

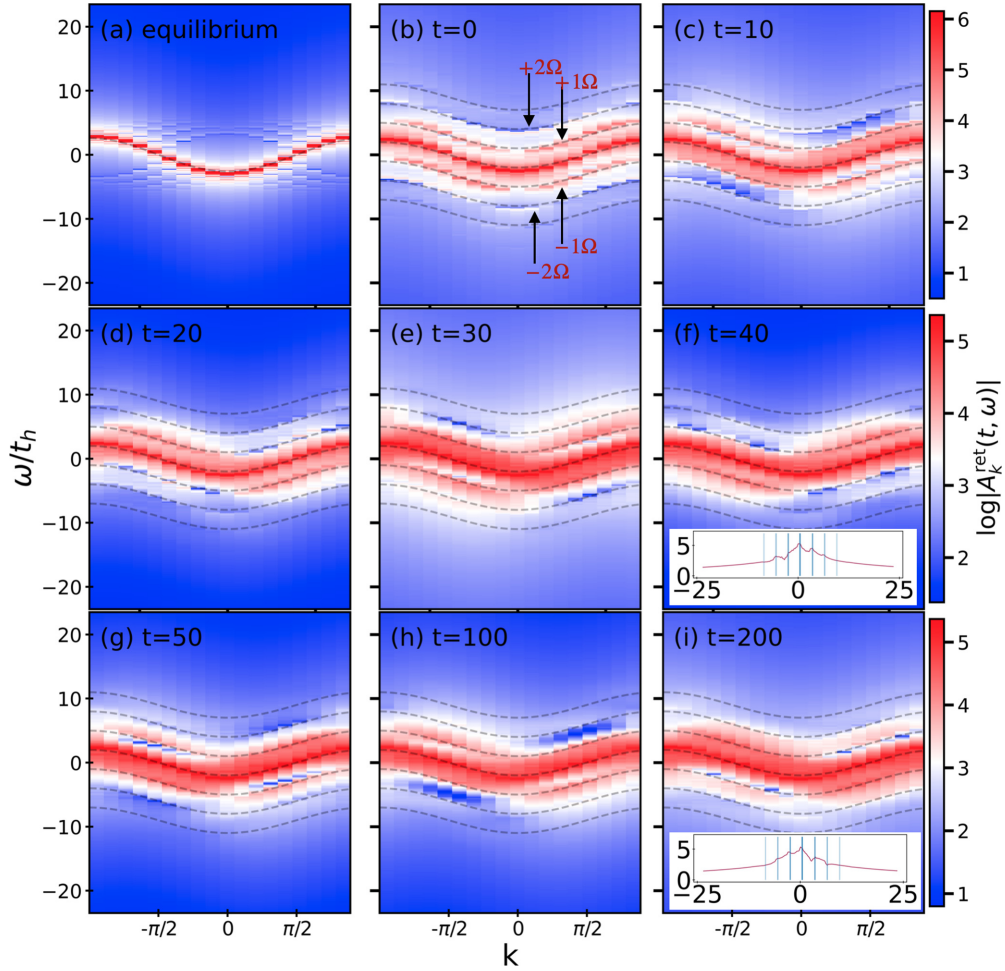


FIG. 7. Retarded time-dependent single-particle spectral function $A_k^{\text{ret}}(t, \omega)$ for $L = 18$, $V/t_h = 1.5$, and PBC at half filling obtained with Lanczos time evolution. (a) shows the equilibrium case and (b) to (i) show different waiting times t indicated on the plots for driving parameters $A = 1.0$ and $\Omega/t_h = 3.0$. Dashed lines are guides to the eye and depict a cosine-like band $\epsilon(k) = -t_{LL} \cos(k)$ and its expected Floquet replica for $t_{LL} = 2.6$ and for the driven case $t_{\text{eff}} = t_{LL} J_0(A_0)$, with J_0 the zeroth Bessel function. Arrows in (b) show the locations of the first two ($\pm n\Omega$) side bands. Insets in (f) and (i) show the cross-section plots for $k = \pi/2$.

$\lesssim 10^{-4}$ at the end of the time evolutions treated. It is very expensive to perform the calculations at substantially higher bond-dimension, so that we refrain from doing so. However, we check for the time-evolution of the energy when varying dt , and find that also in these cases the results for different values of dt are in excellent agreement, see Fig. 14 and the discussion further below.

We also analyzed the errors in the time-evolution of the real-space correlations, where we obtain similar trends (high accuracy for $\Omega = 20$, lower accuracy in the low frequency case). However, at waiting time $t = 0$ in all cases the errors are smaller than the signal size at small times, and for $\Omega = 3$ of comparable value with the value of the correlation functions, which is close to zero there. This confirms the statement about fast loss of coherence in the low frequency case, as discussed in the main text. For a more detailed discussion of the error in the correlation functions, see below. To reach longer times we use the Lanczos time evolution method where the computation of one time step is achieved by projecting the time-evolution operator onto a Lanczos basis for which $|\psi(t)\rangle$ is the initial

state of the iteration procedure. In this way, also for time-dependent Hamiltonians, we compute [78]

$$e^{-idt\hat{H}(t)}|\psi(t)\rangle \approx \mathbf{V}_n(t)e^{-idt\mathbf{T}_n(t)}\mathbf{V}_n^\dagger(t)|\psi(t)\rangle, \quad (\text{A7})$$

where the matrix $\mathbf{V}_n(t)$ contains the Lanczos vectors after n iterations, and $\mathbf{T}_n(t)$ is the tridiagonal matrix representation of $H(t)$ in this basis. As for the MPS calculation, for time-dependent Hamiltonians there are better approximations (see, e.g., Ref. [100]); however, if the value of dt is chosen small enough, we obtain a good accuracy also with this simpler approach. The Lanczos simulations were done using the QuSpin package, see Ref. [61].

APPENDIX B: WINDOW FUNCTION

The retarded single-particle spectral function is computed via

$$A_k^{\text{ret,eq}}(\omega) = -\text{Im} \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} d\tau e^{i\omega\tau} W(\tau) G_{kk}^{\text{ret,eq}}(t, \tau), \quad (\text{B1})$$

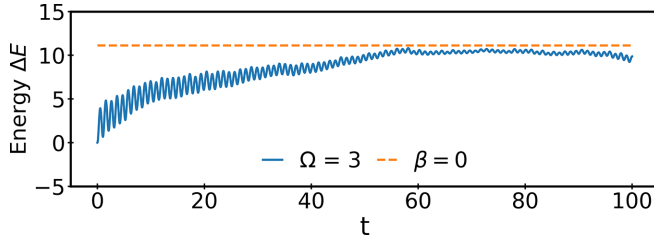


FIG. 8. ED results for the energy gain compared to the initial state (ground state energy E_{GS}), $\Delta E(t) = \langle H(t) \rangle - E_{GS}$ for the driven case with $\Omega = 3.0$ for $L = 14$ and $V/t_h = 1.5$ at half filling and with PBC. The dashed line indicates the result at infinite temperature.

where $W(\tau)$ is a windowing function. Typically, we use an adapted Tukey window [101] with the parameter $\alpha = 0.7$, where

$$W(\tau) = \frac{1}{2} \left[1 - \cos \left(\frac{2\pi\tau}{\alpha N} \right) \right], \text{ for } 0 \leq \tau < \frac{\alpha N}{2}. \quad (\text{B2})$$

In Fig. 6, we show an example of the window function in Eq. (B2) for $\alpha = 0.7$. We find that with this choice even weak FBs can be resolved in the spectral functions shown in the main text.

APPENDIX C: $A_k^{\text{ret}}(t, \omega)$ FOR $V/t_h = 1.5$

We show in Fig. 7 the spectral function $A_k^{\text{ret}}(t, \omega)$ for short to long waiting times, for $L = 18$, $V/t_h = 1.5$, and PBC at half filling, obtained using Lanczos time evolution. In Fig. 7(b), we indicate the Floquet bands (FBs) with arrows to show the locations of the $\pm 1\Omega$ and $\pm 2\Omega$ sidebands. Dashed lines are included in all the figures as guides to the eye. We observe clear FBs for a driving frequency of $\Omega = 3.0$ and $A = 1.0$ for waiting times $t \lesssim 50$. For the chosen driving frequency, overlapping FBs appear. For waiting times $t > 50$, we observe

a significant suppression of FBs and strong spectral intensities in the zeroth Floquet sector, indicating heating effects. This is further illustrated by the insets in Fig. 7, which show the intensity of the FBs at the indicated waiting times and further supported by Fig. 8, where we plot the energy gain as a function of time to monitor the heating.

APPENDIX D: REAL SPACE CORRELATIONS

We show the real and imaginary parts of the real-space correlation function

$$G_{r,L/2}(t, \tau) = \langle \psi(t) | [c_r^\dagger(\tau), c_{L/2}] | \psi(t) \rangle$$

in Fig. 9 for $L = 32$ with OBC for $V/t_h = 5.0$ at waiting time $t = 5.0$ for $\Omega = 3.0$ and at waiting time $t = 20.0$ for $\Omega = 20.0$. The results are presented at equilibrium in panels (a) and (d), for $\Omega = 3.0$ in panels (b) and (e), and for $\Omega = 20.0$ in panels (c) and (f). In equilibrium, we observe a light-cone-like spread of correlations as the system is perturbed at the central site. This behavior changes for $\Omega = 3.0$, as also pointed out in the main text. Already at $t = 0$, we observed a suppression of correlation propagation as from the main text Fig. 5; here, at $t = 5.0$ we observe even stronger suppression. For $\Omega = 20.0$, even at $t = 20.0$, the light-cone-like spread is clearly visible, exhibiting coherence features also at later waiting times.

Furthermore, we compare the imaginary part of $G_{r,L/2}(t, \tau)$ for cases with and without noise in Fig. 10 at longer waiting times. In Fig. 10, we show results for $L = 18$ (PBC), $V/t_h = 5.0$ at half filling. Panels (a) and (b) correspond to the case without noise, while panels (c) and (d) represent the noise case at waiting times $t = 0$ and $t = 100$, respectively. As is evident from the noise case, at longer waiting times the system loses coherence, as seen in the vanishing imaginary part of the correlation $G_{r,L/2}(t, \tau)$ in Fig. 10(d).

In order to estimate the errors in the MPS results for the real time correlation function, we compared results for the coherent drive with $dt = 0.04$ and $dt = 0.005$. For $\Omega = 20$, we

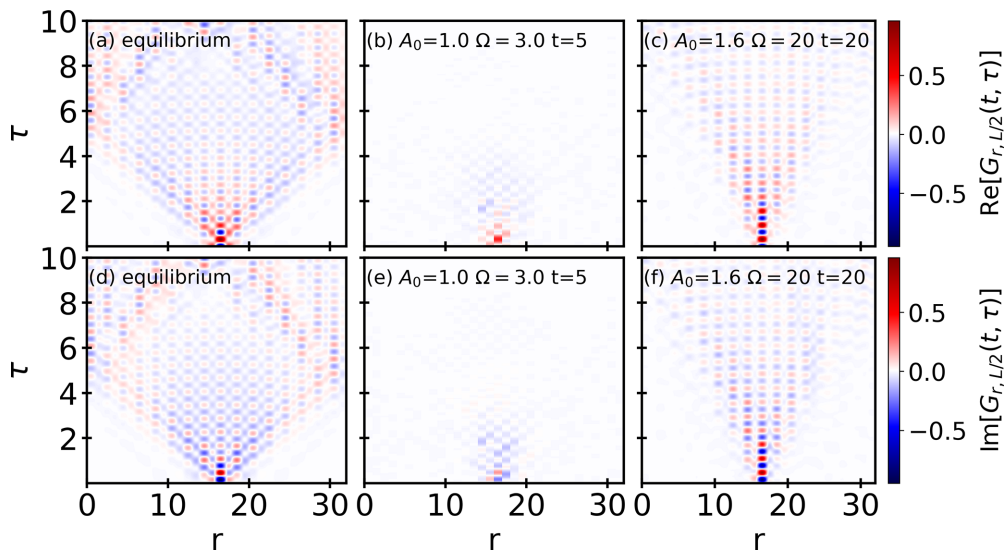


FIG. 9. MPS results for $L = 32$, $V/t_h = 5.0$ at half filling and OBC for the real part (a)–(c) and for the imaginary part (e)–(f) of the real-space correlation function $G_{r,L/2}(t, \tau) = \langle \psi(t) | [c_r^\dagger(\tau), c_{L/2}] | \psi(t) \rangle$ at equilibrium and for the driven case for waiting times $t = 5.0$ and 20.0 .

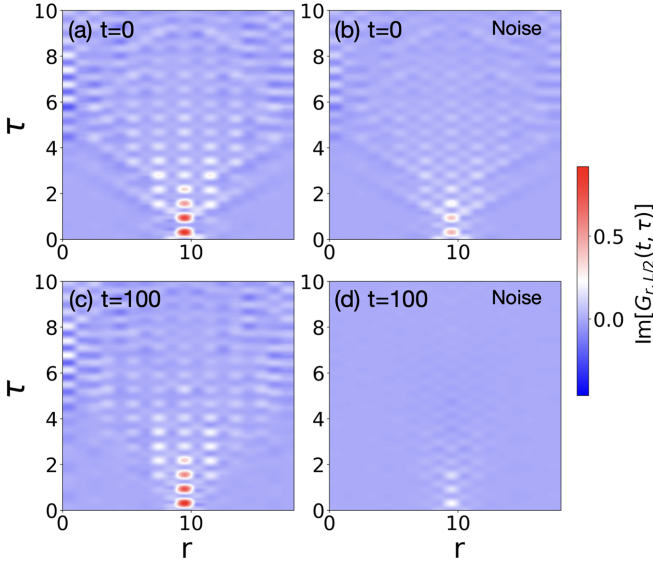


FIG. 10. Lanczos results for $L = 18$ and PBC for the imaginary part of the real-space correlation function $G_{r,L/2}(t, \tau) = \langle \psi(t) | [c_r^\dagger(\tau), c_{L/2}] | \psi(t) \rangle$, with $A_0 = 1.6$ and $\Omega = 20.0$. (a) and (c) show results for waiting time $t = 0$ and $t = 100$ without noise. (b) and (d) show results for waiting time $t = 0$ and $t = 100$ with noise.

find in all cases that the difference between both calculations is substantially smaller than the magnitude of $G_{r,L/2}(t, \tau)$; the difference is (in the case of small values) of the order of a few percent. For $\Omega = 3$, the situation is more complicated, because at later times the value of $G_{r,L/2}(t, \tau)$ becomes very small, so that even small numerical errors lead to a larger relative deviation of the results. However, we find for waiting time $t = 0$ that at short times the difference is $\lesssim 0.015$, while the maximal value of $G_{r,L/2}(t, \tau) \sim 0.16$. At later times, the maximal value of $G_{r,L/2}(t, \tau) \sim 10^{-3}$, and the difference between both runs is of comparable magnitude. Therefore, it is difficult to make quantitative statements at the longer times; however, it is clear that $G_{r,L/2}(t, \tau)$ decayed to a value much smaller than the one at the beginning of the time evolution, clearly indicating the loss of coherence in this case. This is further confirmed by Fig. 11, in which Lanczos results for $\Omega = 3$ for $L = 18$ (PBC) clearly show the loss of coherence

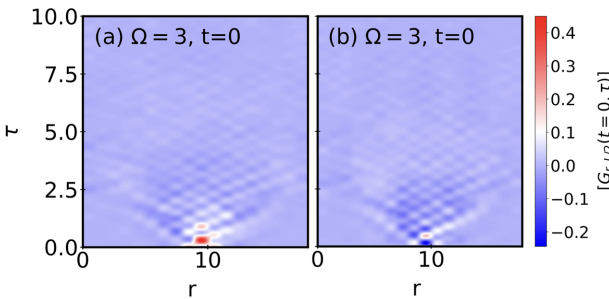


FIG. 11. Lanczos results for $L = 18$ (PBC) for the real-space correlation function $G_{r,L/2}(t, \tau) = \langle \psi(t) | [c_r^\dagger(\tau), c_{L/2}] | \psi(t) \rangle$ at waiting time $t = 0$, with $A_0 = 1.0$ and $\Omega = 3.0$. (a) real part and (b) imaginary part.

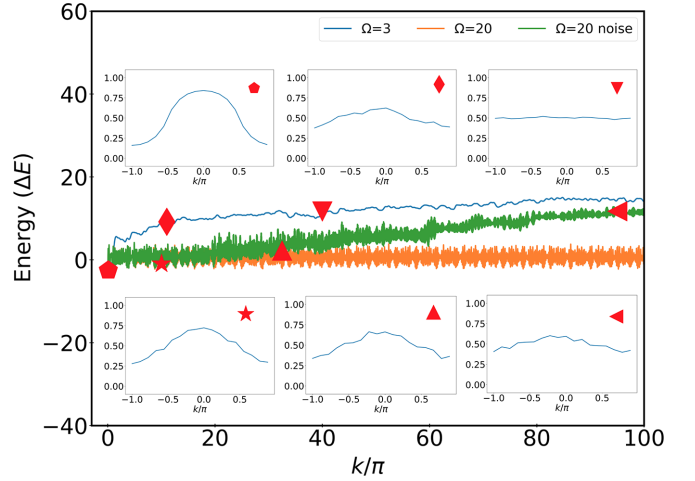


FIG. 12. Time-dependent Lanczos results for systems with $L = 18$ and PBC for the time evolution of the energy $E(t) = \langle H(t) \rangle$ (main figure) for the indicated different driven cases for $dt = 0.005$, and $\langle n_k \rangle(t)$ (insets) for the indicated markers on the energy curves.

at waiting time $t = 0$. At later waiting times, the errors in the MPS calculations for $\Omega = 20$ are similar to the ones at $t = 0$, but larger for $\Omega = 3$ due to the higher entanglement in the system; see Fig. 9 for results at equilibrium and at waiting times $t = 5$ and $t = 20$. However, the results are qualitatively very similar to the ones at waiting time $t = 0$ shown in the main text (see also the discussion above).

APPENDIX E: ENERGY $E(t)$ AND MOMENTUM DISTRIBUTION FUNCTION $\langle n_k \rangle(t)$

In Fig. 12, we present the expectation value $E(t) = \langle H(t) \rangle$ for the driven system with and without noise, comparing the energy gain by plotting the energy for $L = 18$, periodic boundary conditions (PBC), and $V/t_h = 5$ at half filling. We consider driving frequencies $\Omega = 3$, and 20 (both with and without noise) and illustrate the resulting energy gain over time. The insets of Fig. 12 display the momentum distribution function $\langle n_k \rangle(t) = \langle c_k^\dagger c_k \rangle(t)$, with $c_k^{(\dagger)} = 1/\sqrt{L} \sum_{r=1}^L e^{(-)ik \cdot r} c_r^{(\dagger)}$ at the times indicated by the markers. Note that, as we also mention in the main text, $\langle n_k \rangle(t)$ can be related to the Quantum Fisher Information (QFI) $F_Q(k, t)$ for the chosen witness operator $\hat{O} = \sum_i (a_i c_i^\dagger + a_i^* c_i)$. We use $F_Q(k, t) = \int d\omega A(\omega, k, t)$, at temperature $T = 0$ where $A(\omega, k, t) = \text{Im} \int dt' e^{i\omega t'} i \langle [\hat{O}(k, t'), \hat{O}(k, 0)] \rangle$, and integrating over ω results in the momentum distribution function $\langle n_k \rangle(t)$ [56,65,68]. In Fig. 12, the insets show loss of coherence as the system gains energy and the effective temperature increases over time. In the following, we use matrix product states (MPS) [57,58] to compute systems with open boundary conditions (OBC) and pinning field $\mu = V/t_h$ (see main text) for $L = 32$. We employ Lanczos time evolution [78] for systems with $L = 18$ and PBC to achieve longer simulation times than accessible with MPS. In Fig. 13(a) we show MPS results for $\langle n_k \rangle(t)$ for $\Omega = 3.0$ and 20.0 at various waiting times obtained with OBC (note that in the main text and in Fig. 12 we show results obtained with PBC; in Fig. 13(a), we have only positive k -values due to the sine-transform). For

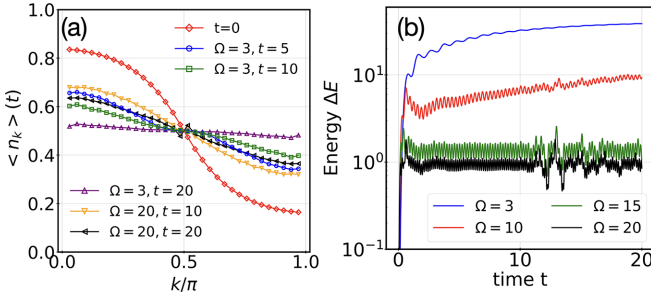


FIG. 13. MPS results for $L = 32$ and $V/t_h = 5.0$ at half filling and with OBC: (a) momentum distribution function $\langle n_k \rangle(t)$ at different waiting times for low $\Omega/t_h = 3.0$ and high frequency $\Omega/t_h = 20.0$. (b) energy $\Delta E(t) = \langle H(t) \rangle - E_{GS}$ as a function of time for the driven case with the different frequencies as indicated.

$\Omega = 3.0$, we observe a strong flattening of the momentum distribution function, consistent with the insets of Fig. 12. Further, in Fig. 13(b) we check $E(t)$ for $L = 32$ using MPS for different values of Ω , which shows that the energy gain decreases with increase of the driving frequency. For the case of incoherent noise in the driving frequency, we consider a driving field with vector potential $A_V(t) = A_0 \sin((\Omega \pm \Omega_{\text{noise}})t)$, where the frequency is given by $\Omega = 20 \pm \Omega_{\text{noise}}$, $\Omega_{\text{noise}} \in [0, 0.1]$. We discretize the time as $t = n dt$ with $n \in \mathbb{N}_0$ and apply $A_V(n dt) = A_0 \sin((\Omega \pm \Omega_{\text{noise}})n dt)$. Therefore, at each time step of the simulation, we have a different value of Ω_{noise} . Hence, the frequency, with which the value of Ω_{noise}

is changed, depends on the value of dt , so that we can expect different time-dependent behavior of the observables for different values of dt in the presence of noise. In Fig. 14(a), we show the energy gain $E(t)$ for discrete time steps $dt = 0.05$ and $dt = 0.005$ for $L = 14$ PBC. We observe that the choice of dt affects the energy gain over time in the presence of noise, as mentioned above.

Furthermore, we check the dependence of our results on the size of dt also without noise in the driving. This is important, since we are treating a time-dependent Hamiltonian, and in addition to the usual MPS errors, we have an additional source of errors due to the discretization of t in $H(t)$ [100]. In Fig. 14(a) we show Lanczos results for the different cases treated in the main text, but with values of $dt = 0.05$ or $dt = 0.005$, respectively, in order to estimate the error due to the discretization of $H(t)$. As can be seen, in all cases the values of the energy $E(t)$ lie on top of each other, so that this discretization error seems to play a minor role. In Fig. 14(b) we turn to MPS results for $L = 32$ (OBC) for $\Omega = 3.0$ and in Fig. 14(c) for $\Omega = 20.0$, with discrete time steps $dt = 0.04$, $dt = 0.005$, and $dt = 0.0025$. In these plots, the MPS results all carry the accumulated error due to (i) discretization of time in $H(t)$; (ii) error of the TDVP-ansatz due to the finite value of dt [58]; (iii) accumulated MPS-errors due to the finite bond dimension [57]. Hence, by comparing the time evolution of observables with different values of dt , we get an estimate for the accumulated errors of our numerical calculations, which can be used to estimate the overall accuracy of our results. As can be seen, only the values with the largest chosen value of

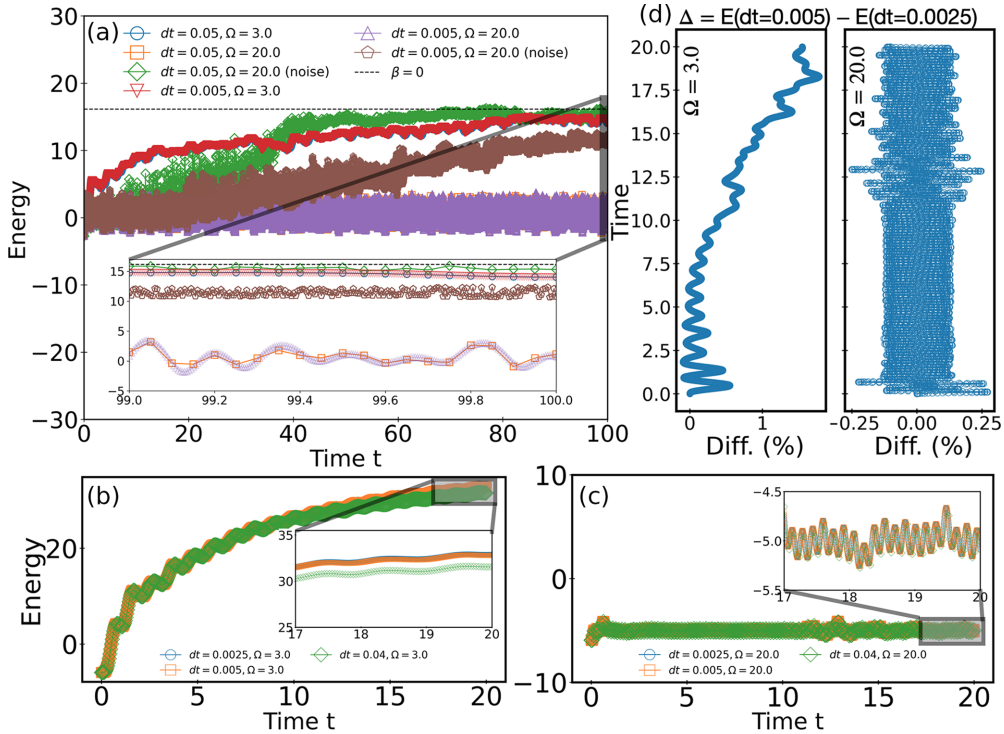


FIG. 14. Energy $E(t) = \langle H(t) \rangle$ as a function of time for $V/t_h = 5.0$. (a) Lanczos results for the time evolution of $E(t)$ for the driven case for different values of Ω for $L = 14$ (PBC) when using $dt = 0.05$, or $dt = 0.005$, respectively. The inset is a zoom into the interval $t = [99, 100.0]$. MPS results for $E(t)$ in (b) for the driven case of $\Omega = 3.0$ and in (c) for the driven case of $\Omega = 20.0$ using $dt = 0.04$, $dt = 0.005$, and $dt = 0.0025$. The insets in (b) and (c) show close-ups of the interval $t = [17, 20.0]$. (d) relative difference for the energy for the choice of $dt = 0.005$ and $dt = 0.0025$, respectively.

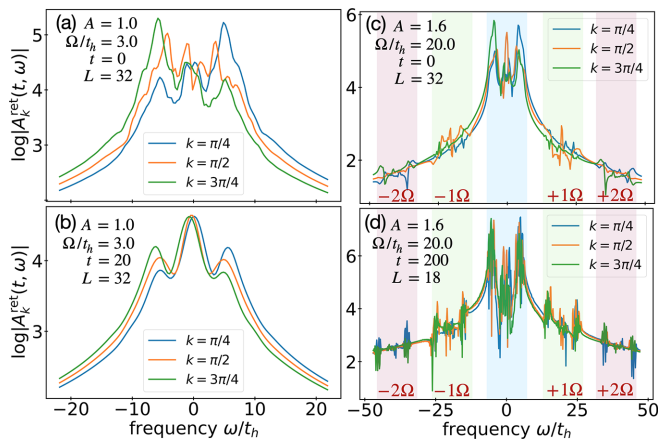


FIG. 15. Results for $A_k^{\text{ret}}(t, \omega)$ for $V/t_h = 5.0$ at half filling at different values of k . (a) and (b) show the results for the driven case with $L = 32$, $A = 1.0$, and $\Omega = 3.0$ at waiting time $t = 0$ and $t = 20$, respectively. (c) shows the plots for the driven case with $L = 32$, $A = 1.6$, and $\Omega = 20.0$ for waiting time $t = 0$. (d) shows the plots for the driven case with $L = 18$, $A = 1.6$, and $\Omega = 20.0$ for waiting time $t = 200$. The shaded regions show the location of the FBs as a guide to the eye at $\pm n\Omega$ in the panels (c) and (d).

$dt = 0.04$ for $\Omega = 3$ show a clear discrepancy at later times, indicating that the results obtained with $dt = 0.005$ shown in the main text have a high numerical accuracy. In Fig. 14(d) this is further quantified by showing the relative difference (in

percentage) for the energy $E(t)$ with the choice of $dt = 0.005$, and $dt = 0.0025$. For the coherent drive case, the maximal difference of these results at the end of the time evolutions shown is $\approx 2\%$ for $\Omega = 3$ and $\approx 0.25\%$ for $\Omega = 20$. The smaller error in the high frequency case can be understood by having less heating and less growth of entanglement in this case (this can be seen in the growth of the discarded weight with time, which is much faster in the low frequency case, see further below). We, therefore, stick to the discrete time steps of $dt = 0.005$ for all our calculations presented in the main text.

APPENDIX F: FURTHER RESULTS AT FIXED k

In Fig. 15 we show further results of $A_k^{\text{ret}}(t, \omega)$ when keeping k fixed and at different values of k for $V/t_h = 5.0$, see Figs. 2 and 3 in the main text. In Figs. 15(a) and 15(b) we show the results for $L = 32$ (MPS), $A = 1.0$, and $\Omega = 3.0$ for waiting time $t = 0$ and $t = 20$. As we have mentioned in the main text, we observe high intensity in the gap region at finite waiting time $t = 20$ across different momenta as shown in the plots. Figure 15(c) shows the results for $L = 32$ (MPS), $A = 1.6$, and $\Omega = 20.0$ for waiting time $t = 0$. We observe the intensities of the sidebands at $\pm n\Omega$ across different momenta shown in the plot. Furthermore, we also show results at longer waiting time, $t = 200$, in (d) for $L = 18$, $A = 1.6$, and $\Omega = 20.0$. We again observe stable FBs at the different values of k shown here. We highlight by shaded regions as guides to the eye the location of the FBs in the (c) and (d) subplots.

- [1] D. N. Basov, R. D. Averitt, and D. Hsieh, Towards properties on demand in quantum materials, *Nat. Mater.* **16**, 1077 (2017).
- [2] T. Oka and S. Kitamura, Floquet engineering of quantum materials, *Annu. Rev. Condens. Matter Phys.* **10**, 387 (2019).
- [3] A. de la Torre, D. M. Kennes, M. Claassen, S. Gerber, J. W. McIver, and M. A. Sentef, Colloquium: Nonthermal pathways to ultrafast control in quantum materials, *Rev. Mod. Phys.* **93**, 041002 (2021).
- [4] C. Bao, P. Tang, D. Sun, and S. Zhou, Light-induced emergent phenomena in 2D materials and topological materials, *Nat. Rev. Phys.* **4**, 33 (2021).
- [5] V. N. Valmispild, E. Gorelov, M. Eckstein, A. I. Lichtenstein, H. Aoki, M. I. Katsnelson, M. Y. Ivanov, and O. Smirnova, Sub-cycle multidimensional spectroscopy of strongly correlated materials, *Nat. Photon.* **18**, 432 (2024).
- [6] J. Li and M. Eckstein, Manipulating intertwined orders in solids with quantum light, *Phys. Rev. Lett.* **125**, 217402 (2020).
- [7] Y. Murakami, D. Golež, M. Eckstein, and P. Werner, Photo-induced nonequilibrium states in Mott insulators, *Rev. Mod. Phys.* **97**, 035001 (2025).
- [8] A. G. Grushin, A. Gómez-León, and T. Neupert, Floquet fractional Chern insulators, *Phys. Rev. Lett.* **112**, 156801 (2014).
- [9] G. Jotzu, M. Messer, R. Desbuquois, M. Lebrat, T. Uehlinger, D. Greif, and T. Esslinger, Experimental realization of the topological Haldane model with ultracold fermions, *Nature (London)* **515**, 237 (2014).
- [10] T. Mikami, S. Kitamura, K. Yasuda, N. Tsuji, T. Oka, and H. Aoki, Brillouin-Wigner theory for high-frequency expansion in periodically driven systems: Application to Floquet topological insulators, *Phys. Rev. B* **93**, 144307 (2016).
- [11] N. H. Lindner, G. Refael, and V. Galitski, Floquet topological insulator in semiconductor quantum wells, *Nat. Phys.* **7**, 490 (2011).
- [12] J. W. McIver, B. Schulte, F.-U. Stein, T. Matsuyama, G. Jotzu, G. Meier, and A. Cavalleri, Light-induced anomalous Hall effect in graphene, *Nat. Phys.* **16**, 38 (2020).
- [13] H. Dehghani, T. Oka, and A. Mitra, Out-of-equilibrium electrons and the Hall conductance of a Floquet topological insulator, *Phys. Rev. B* **91**, 155422 (2015).
- [14] H. Dehghani and A. Mitra, Optical Hall conductivity of a Floquet topological insulator, *Phys. Rev. B* **92**, 165111 (2015).
- [15] T. Oka and H. Aoki, Photovoltaic Hall effect in graphene, *Phys. Rev. B* **79**, 081406(R) (2009).
- [16] I. Bloch, J. Dalibard, and W. Zwerger, Many-body physics with ultracold gases, *Rev. Mod. Phys.* **80**, 885 (2008).
- [17] K. Wintersperger, M. Bukov, J. Näger, S. Lellouch, E. Demler, U. Schneider, I. Bloch, N. Goldman, and M. Aidelsburger, Parametric instabilities of interacting bosons in periodically driven 1D optical lattices, *Phys. Rev. X* **10**, 011030 (2020).
- [18] C. Weitenberg and J. Simonet, Tailoring quantum gases by Floquet engineering, *Nat. Phys.* **17**, 1342 (2021).
- [19] M. Lei, R. Fukumori, C.-J. Wu, E. Barnes, S. Economou, J. Choi, and A. Faraon, Quantum thermalization and Floquet

- engineering in a spin ensemble with a clock transition, *Nat. Phys.* **21**, 1196 (2025).
- [20] A. Di Carli, R. Cruickshank, M. Mitchell, A. La Rooij, S. Kuhr, C. E. Creffield, and E. Haller, Instabilities of interacting matter waves in optical lattices with Floquet driving, *Phys. Rev. Res.* **5**, 033024 (2023).
- [21] A. Eckardt, Colloquium: Atomic quantum gases in periodically driven optical lattices, *Rev. Mod. Phys.* **89**, 011004 (2017).
- [22] M. Sentef, M. Claassen, A. Kemper, B. Moritz, T. Oka, J. Freericks, and T. Devereaux, Theory of Floquet band formation and local pseudospin textures in pump-probe photoemission of graphene, *Nat. Commun.* **6**, 7047 (2015).
- [23] Y. H. Wang, H. Steinberg, P. Jarillo-Herrero, and N. Gedik, Observation of Floquet-Bloch states on the surface of a topological insulator, *Science* **342**, 453 (2013).
- [24] F. Mahmood, C.-K. Chan, Z. Alpichshev, D. Gardner, Y. Lee, P. A. Lee, and N. Gedik, Selective scattering between Floquet-Bloch and Volkov states in a topological insulator, *Nat. Phys.* **12**, 306 (2016).
- [25] U. De Giovannini, H. Hübener, and A. Rubio, Monitoring electron-photon dressing in WSe₂, *Nano Lett.* **16**, 7993 (2016).
- [26] S. Aeschlimann, S. A. Sato, R. Krause, M. Chávez-Cervantes, U. De Giovannini, H. Hübener, S. Forti, C. Coletti, K. Hanff, K. Rossnagel, A. Rubio, and I. Gierz, Survival of Floquet-Bloch states in the presence of scattering, *Nano Lett.* **21**, 5028 (2021).
- [27] M. Merboldt, M. Schüler, D. Schmitt, J. P. Bange, W. Bennecke, K. Gadge, K. Pierz, H. W. Schumacher, D. Momeni, D. Steil, S. R. Manmana, M. A. Sentef, M. Reutz, and S. Mathias, Observation of Floquet states in graphene, *Nat. Phys.* **21**, 1093 (2025).
- [28] D. Choi, M. Mogi, U. De Giovannini, D. Azoury, B. Lv, Y. Su, H. Hübener, A. Rubio, and N. Gedik, Observation of Floquet-Bloch states in monolayer graphene, *Nat. Phys.* **21**, 1100 (2025).
- [29] N. Bielinski, R. Chari, J. May-Mann, S. Kim, J. Zwettler, Y. Deng, A. Aishwarya, S. Roychowdhury, C. Shekhar, M. Hashimoto, D. Lu, J. Yan, C. Felser, V. Madhavan, Z.-X. Shen, T. L. Hughes, and F. Mahmood, Floquet-Bloch manipulation of the Dirac gap in a topological antiferromagnet, *Nat. Phys.* **21**, 458 (2025).
- [30] S. Fragkos, B. Fabre, O. Tkach, S. Petit, D. Descamps, G. Schönhense, Y. Mairesse, M. Schüler, and S. Beaulieu, Floquet-Bloch valleytronics, *Nat. Commun.* **16**, 5799 (2025).
- [31] M. Schüler, U. De Giovannini, H. Hübener, A. Rubio, M. A. Sentef, T. P. Devereaux, and P. Werner, How circular dichroism in time- and angle-resolved photoemission can be used to spectroscopically detect transient topological states in graphene, *Phys. Rev. X* **10**, 041013 (2020).
- [32] M. Schüler and S. Beaulieu, Probing topological Floquet states in WSe₂ using circular dichroism in time- and angle-resolved photoemission spectroscopy, *Commun. Phys.* **5**, 164 (2022).
- [33] Y. Murakami and P. Werner, Nonequilibrium steady states of electric field driven Mott insulators, *Phys. Rev. B* **98**, 075102 (2018).
- [34] N. Tsuji, T. Oka, and H. Aoki, Correlated electron systems periodically driven out of equilibrium: Floquet+DMFT formalism, *Phys. Rev. B* **78**, 235124 (2008).
- [35] H. Aoki, N. Tsuji, M. Eckstein, M. Kollar, T. Oka, and P. Werner, Nonequilibrium dynamical mean-field theory and its applications, *Rev. Mod. Phys.* **86**, 779 (2014).
- [36] M. Puviani and F. Manghi, Periodically driven interacting electrons in one dimension: Many-body Floquet approach, *Phys. Rev. B* **94**, 161111(R) (2016).
- [37] D. H. Dunlap and V. M. Kenkre, Dynamic localization of a charged particle moving under the influence of an electric field, *Phys. Rev. B* **34**, 3625 (1986).
- [38] A. Osterkorn, C. Meyer, and S. R. Manmana, In-gap band formation in a periodically driven charge density wave insulator, *Commun. Phys.* **6**, 245 (2023).
- [39] P. Ponte, Z. Papić, F. Huveneers, and D. A. Abanin, Many-body localization in periodically driven systems, *Phys. Rev. Lett.* **114**, 140401 (2015).
- [40] A. Lazarides, A. Das, and R. Moessner, Fate of many-body localization under periodic driving, *Phys. Rev. Lett.* **115**, 030402 (2015).
- [41] P. Sierant, M. Lewenstein, A. Scardicchio, and J. Zakrzewski, Stability of many-body localization in Floquet systems, *Phys. Rev. B* **107**, 115132 (2023).
- [42] L. Zhang, V. Khemani, and D. A. Huse, A Floquet model for the many-body localization transition, *Phys. Rev. B* **94**, 224202 (2016).
- [43] K. S. C. Decker, C. Karrasch, J. Eisert, and D. M. Kennes, Floquet engineering topological many-body localized systems, *Phys. Rev. Lett.* **124**, 190601 (2020).
- [44] M. Bukov, L. D'Alessio, and A. Polkovnikov, Universal high-frequency behavior of periodically driven systems: from dynamical stabilization to Floquet engineering, *Adv. Phys.* **64**, 139 (2015).
- [45] D. Kennes, A. de la Torre, A. Ron, D. Hsieh, and A. Millis, Floquet engineering in quantum chains, *Phys. Rev. Lett.* **120**, 127601 (2018).
- [46] L. Broers and L. Mathey, Observing light-induced Floquet band gaps in the longitudinal conductivity of graphene, *Commun. Phys.* **4**, 248 (2021).
- [47] L. Broers and L. Mathey, Detecting light-induced Floquet band gaps of graphene via trARPES, *Phys. Rev. Res.* **4**, 013057 (2022).
- [48] N. Tsuji, in *Encyclopedia of Condensed Matter Physics* (2nd Ed.), edited by T. Chakraborty (Academic Press, Oxford, 2024), pp. 967–980.
- [49] C. Fleckenstein and M. Bukov, Prethermalization and thermalization in periodically driven many-body systems away from the high-frequency limit, *Phys. Rev. B* **103**, L140302 (2021).
- [50] M. Bukov, M. Heyl, D. A. Huse, and A. Polkovnikov, Heating and many-body resonances in a periodically driven two-band system, *Phys. Rev. B* **93**, 155132 (2016).
- [51] H. Zhao, J. Knolle, R. Moessner, and F. Mintert, Suppression of interband heating for random driving, *Phys. Rev. Lett.* **129**, 120605 (2022).
- [52] D. A. Abanin, W. De Roeck, and F. Huveneers, Exponentially slow heating in periodically driven many-body systems, *Phys. Rev. Lett.* **115**, 256803 (2015).
- [53] A. Haldar, R. Moessner, and A. Das, Onset of Floquet thermalization, *Phys. Rev. B* **97**, 245122 (2018).

- [54] F. Peronaci, M. Schiró, and O. Parcollet, Resonant thermalization of periodically driven strongly correlated electrons, *Phys. Rev. Lett.* **120**, 197601 (2018).
- [55] F. Machado, G. D. Kahanamoku-Meyer, D. V. Else, C. Nayak, and N. Y. Yao, Exponentially slow heating in short and long-range interacting Floquet systems, *Phys. Rev. Res.* **1**, 033202 (2019).
- [56] P. Hauke, M. Heyl, L. Tagliacozzo, and P. Zoller, Measuring multipartite entanglement through dynamic susceptibilities, *Nat. Phys.* **12**, 778 (2016).
- [57] U. Schollwöck, The density-matrix renormalization group in the age of matrix product states, *Ann. Phys.* **326**, 96 (2011).
- [58] S. Paeckel, T. Köhler, A. Swoboda, S. R. Manmana, U. Schollwöck, and C. Hubig, Time-evolution methods for matrix-product states, *Ann. Phys.* **411**, 167998 (2019).
- [59] R. M. Noack and S. R. Manmana, Diagonalization- and numerical renormalization-group-based methods for interacting quantum systems, *AIP Conf. Proc.* **789**, 93 (2005).
- [60] A. W. Sandvik, Computational studies of quantum spin systems, *AIP Conf. Proc.* **1297**, 135 (2010).
- [61] P. Weinberg and M. Bukov, QuSpin: a Python package for dynamics and exact diagonalisation of quantum many body systems. part II: bosons, fermions and higher spins, *SciPost Phys.* **7**, 020 (2019).
- [62] A. Scheie, P. Laurell, B. Lake, S. E. Nagler, M. B. Stone, J.-S. Caux, and D. A. Tennant, Quantum wake dynamics in Heisenberg antiferromagnetic chains, *Nat. Commun.* **13**, 5796 (2022).
- [63] A. Scheie, P. Laurell, A. M. Samarakoon, B. Lake, S. E. Nagler, G. E. Granroth, S. Okamoto, G. Alvarez, and D. A. Tennant, Witnessing entanglement in quantum magnets using neutron scattering, *Phys. Rev. B* **103**, 224434 (2021).
- [64] A. Scheie, P. Laurell, W. Simeth, E. Dagotto, and D. A. Tennant, Tutorial: Extracting entanglement signatures from neutron spectroscopy, *Materials Today Quantum* **5**, 100020 (2025).
- [65] F. Mazza, S. Biswas, X. Yan, A. Prokofiev, P. Steffens, Q. Si, F. F. Assaad, and S. Paschen, Quantum Fisher information in a strange metal, *arXiv:2403.12779*.
- [66] R. K. Malla, A. Weichselbaum, T.-C. Wei, and R. M. Konik, Detecting multipartite entanglement patterns using single particle Green's functions, *Phys. Rev. Lett.* **133**, 260202 (2024).
- [67] J. Hales, U. Bajpai, T. Liu, D. R. Baykusheva, M. Li, M. Mitrano, and Y. Wang, Witnessing light-driven entanglement using time-resolved resonant inelastic x-ray scattering, *Nat. Commun.* **14**, 3512 (2023).
- [68] D. R. Baykusheva, M. H. Kalthoff, D. Hofmann, M. Claassen, D. M. Kennes, M. A. Sentef, and M. Mitrano, Witnessing nonequilibrium entanglement dynamics in a strongly correlated fermionic chain, *Phys. Rev. Lett.* **130**, 106902 (2023).
- [69] R. Peierls, Zur theorie des diamagnetismus von leitungselektronen, *Z. Phys.* **80**, 763 (1933).
- [70] T. Giamarchi, *Quantum Physics in One Dimension, International Series of Monographs on Physics* (Oxford, Oxford Academic, 2007).
- [71] J. D. Cloizeaux and M. Gaudin, Anisotropic linear magnetic chain, *J. Math. Phys.* **7**, 1384 (1966).
- [72] C. Meyer, Matrix product state approaches to non-equilibrium spectral quantities of strongly correlated fermions in one dimension, Ph.D. thesis, Goettingen University, Goettingen, 2022.
- [73] M. H. Kalthoff, G. S. Uhrig, and J. K. Freericks, Emergence of Floquet behavior for lattice fermions driven by light pulses, *Phys. Rev. B* **98**, 035138 (2018).
- [74] H. T. M. Nghiem and T. A. Costi, Time evolution of the Kondo resonance in response to a quench, *Phys. Rev. Lett.* **119**, 156601 (2017).
- [75] H. T. M. Nghiem, H. T. Dang, and T. A. Costi, Time-dependent spectral functions of the Anderson impurity model in response to a quench with application to time-resolved photoemission spectroscopy, *Phys. Rev. B* **101**, 115117 (2020).
- [76] G. S. Uhrig, M. H. Kalthoff, and J. K. Freericks, Positivity of the spectral densities of retarded Floquet Green functions, *Phys. Rev. Lett.* **122**, 130604 (2019).
- [77] T. Blum, R. M. Noack, and S. R. Manmana, Time evolution of the local density of states of strongly correlated fermions coupled to a nanoprobe, *Phys. Rev. B* **111**, 035152 (2025).
- [78] S. R. Manmana, A. Muramatsu, and R. M. Noack, Time evolution of one-dimensional quantum many body systems, *AIP Conf. Proc.* **789**, 269 (2005).
- [79] R. G. Pereira, S. R. White, and I. Affleck, Spectral function of spinless fermions on a one-dimensional lattice, *Phys. Rev. B* **79**, 165113 (2009).
- [80] P. Ponte, A. Chandran, Z. Papić, and D. A. Abanin, Periodically driven ergodic and many-body localized quantum systems, *Ann. Phys.* **353**, 196 (2015).
- [81] R. Moessner and S. L. Sondhi, Equilibration and order in quantum Floquet matter, *Nat. Phys.* **13**, 424 (2017).
- [82] P. Bordia, H. Lüschen, U. Schneider, M. Knap, and I. Bloch, Periodically driving a many-body localized quantum system, *Nat. Phys.* **13**, 460 (2017).
- [83] D. A. Abanin, W. De Roeck, W. W. Ho, and F. Huveneers, Effective Hamiltonians, prethermalization, and slow energy absorption in periodically driven many-body systems, *Phys. Rev. B* **95**, 014112 (2017).
- [84] S. T. Park, Interference in Floquet-Volkov transitions, *Phys. Rev. A* **90**, 013420 (2014).
- [85] E. H. Lieb and D. W. Robinson, The finite group velocity of quantum spin systems, *Commun. Math. Phys.* **28**, 251 (1972).
- [86] P. Calabrese and J. Cardy, Time dependence of correlation functions following a quantum quench, *Phys. Rev. Lett.* **96**, 136801 (2006).
- [87] P. Calabrese and J. Cardy, Quantum quenches in extended systems, *J. Stat. Mech.: Theory Exp.* **2007**, P06008 (2007).
- [88] S. Bravyi, M. B. Hastings, and F. Verstraete, Lieb-Robinson bounds and the generation of correlations and topological quantum order, *Phys. Rev. Lett.* **97**, 050401 (2006).
- [89] A. M. Läuchli and C. Kollath, Spreading of correlations and entanglement after a quench in the one-dimensional Bose-Hubbard model, *J. Stat. Mech.: Theory Exp.* **2008**, P05018 (2008).
- [90] S. R. Manmana, S. Wessel, R. M. Noack, and A. Muramatsu, Time evolution of correlations in strongly interacting fermions after a quantum quench, *Phys. Rev. B* **79**, 155104 (2009).

- [91] M. Cheneau, P. Barmettler, D. Poletti, M. Endres, P. Schauß, T. Fukuhara, C. Gross, I. Bloch, C. Kollath, and S. Kuhr, Light-cone-like spreading of correlations in a quantum many-body system, *Nature (London)* **481**, 484 (2012).
- [92] M. H. Kalthoff, D. M. Kennes, and M. A. Sentef, Floquet-engineered light-cone spreading of correlations in a driven quantum chain, *Phys. Rev. B* **100**, 165125 (2019).
- [93] C. Fleckenstein and M. Bukov, Thermalization and prethermalization in periodically kicked quantum spin chains, *Phys. Rev. B* **103**, 144307 (2021).
- [94] E. Dagotto, Correlated electrons in high-temperature superconductors, *Rev. Mod. Phys.* **66**, 763 (1994).
- [95] K. Gadge, and S. R. Manmana, Stability of Floquet sidebands and quantum coherence in 1D strongly interacting spinless fermions [Data set], Zenodo (2025), <https://doi.org/10.5281/zenodo.15115513>.
- [96] H. Benthien, F. Gebhard, and E. Jeckelmann, Spectral function of the one-dimensional Hubbard model away from half filling, *Phys. Rev. Lett.* **92**, 256401 (2004).
- [97] A. C. Tiegel, Finite-temperature dynamics of low-dimensional quantum systems with DMRG methods, Ph.D. thesis, Goettingen University, Goettingen, 2016.
- [98] T. Köhler, S. Paeckel, C. Meyer, and S. R. Manmana, Formation of spatial patterns by spin-selective excitations of interacting fermions, *Phys. Rev. B* **102**, 235166 (2020).
- [99] S. Paeckel and T. Köhler, Symmps toolkit — symmps.eu, <https://www.symmps.eu>.
- [100] A. Alvermann and H. Fehske, High-order commutator-free exponential time-propagation of driven quantum systems, *J. Comput. Phys.* **230**, 5930 (2011).
- [101] P. Bloomfield, *Fourier Analysis of Time Series*, Wiley Series in Probability and Statistics (John Wiley and Sons, Nashville, TN, 2000).